

Stephen Nguyen

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EDUCATION

Iowa State University

Ames, IA

B.S. in Computer Science and Data Science (GPA: 3.5 out of 4.0)

Expected Dec 2027

- Relevant Coursework:** Data Structures, Algorithms & Complexity; Database Management; System Design; Cloud Computing; Software Testing; Machine Learning.

EXPERIENCE

Bayer HealthCare LLC

May 2026 – Aug 2026

Embedded Software Engineering Co-op

Indianola, PA

- Developed **C and C++ embedded software** for **Cevego Pro injector subsystems**, coordinating piston motor control, injection safety checks, device-state monitoring, and operator GUI workflows across **6 software subsystems**.
- Implemented communication logic across **MCP, ICP, SCP, ISI, and display components**, supporting **2 motor-control processors** for **Axis A and Axis B** and validating injector commands, safety states, and device events.
- Integrated embedded-device workflows with **SPI, I2C, UART, RS-422, MongoDB, and backend services**, improving how injector status, patient-case records, and long-term device data moved across the system.
- Improved validation and debugging for connected medical-device workflows serving **2,000+ connected-device users** by strengthening **state-machine behavior, event logging, watchdog checks, and health signals**, reducing setup and deployment time about **50%**.

Finbud AI

Sep 2025 – Dec 2025

Software Development Intern

Chicago, IL

- Refactored **20+ microservices on Amazon EKS**; unified ingress and service routing with **NGINX and Istio traffic policies**, cutting downtime **70%** (12 hours monthly) for **10K+ users**.
- Reduced MTTR 60%** by turning **Prometheus alerts** into auto-mitigation that triggers **Kafka repartition** at 70% CPU, keeping analytics pipelines stable at **8K+ RPS** under load.
- Lowered peak 5xx 35%** by rolling out **distributed rate limiting and circuit breakers** backed by **AWS ElastiCache (Redis)**, preventing overload cascades and protecting critical endpoints.
- Hardened delivery using **Jenkins pipelines and Helm releases** with **Docker builds and automated k6 gates (80%+ coverage)**, cutting incidents **10%**; improved **Nuxt.js UX via SWR**, bundle-35%, sessions +15%.

Techcombank (Financial Services and Digital Banking)

January 2025 – May 2025

Software Engineer Intern, Backend & Infrastructure

Hanoi, Vietnam

- Engineered **Face ID biometric authentication** with **JWT, OAuth2, and liveness detection**, verifying logins in **under 2s** and securing **5K+ customers and 10K+ transactions** to enable branch-free, fully in-app banking.
- Built an interest-calculation microservice with **Java and Spring Boot** to compute daily accrual and compounding across savings and loan products for **5K+ accounts**, backed by **optimized SQL** to cut interest-posting time **90%** and remove manual branch work.
- Developed **self-service REST APIs** that diagnose and auto-resolve **70%** of common account and technical issues in-app via guided workflows, cutting branch visits **40%** for **5K+ customers**.
- Containerized microservices with **Docker on Kubernetes** and **GitLab continuous integration and delivery**, adding **Redis caching, Kafka streaming, and Prometheus monitoring** on Linux to sustain **10K+ daily transactions at p95 under 250 ms and 99.9% uptime**.

FPT Corporation

May 2024– Aug 2024

Software Engineer Intern

Da Nang, VN

- Delivered a **full-stack e-commerce platform** with **Next.js and MongoDB**, implementing browse, cart, and checkout workflows that sustained **6K+ requests per day** at peak traffic with stable sessions.
- Optimized **storefront page-load performance** in **Next.js**, cutting latency by **~30%** via **image optimization, server-side caching, and critical render-path tuning** on landing and product pages.
- Hardened the **Stripe Webhooks pipeline** with **event persistence and idempotency checks**, deduplicating events and applying **retries with backoff** to prevent duplicate processing under burst traffic.
- Automated release gates with **GitLab CI/CD** and load testing using **k6**, validating **2K+ RPS** while maintaining **80% coverage** and reducing post-release bugs by **15%**.

Iowa State University

Jan 2024– May 2024

Undergraduate Teaching Assistant, Software Development Project Course

Ames, IA

- Mentored **~30 students weekly** building full-stack project applications, guiding **Spring Boot and WebSocket** backends alongside **HTML, JavaScript, and CSS** frontends, with hands-on debugging on real-time features, REST integration, and client-server data flow.
- Reviewed student codebases for **component structure, organization, and correctness**, and automated **edge-case grading scripts** that standardized feedback across teams and cut turnaround time by **40%**.

PROJECTS

WorkApp | Java, Spring Boot, MySQL, WebSocket, GitLab CI and CD

August 2024

- Built an **HR management system** in **Spring Boot** with **40+ MySQL tables** and role-based access control (**RBAC**), applying object-oriented design to reduce data inconsistencies **50%**.
- Shipped a **real-time WebSocket chat interface** and **40+ REST endpoints** behind **GitLab CI and CD quality gates**, handling live message state and role-based views across the application.

ShopArround | Next.js, React, TypeScript, Node.js, Express, AWS, Docker, MongoDB

January 2025

- Built the **storefront frontend** in **Next.js and React** with reusable, responsive components for browse, product, and checkout flows, structured with a **modular component architecture and custom hooks** for maintainability and consistent state handling.
- Cut initial **page-load time ~40%** using **Next.js server-side rendering, image optimization, and code splitting**, keeping product and landing pages fast and responsive under peak traffic.

TECHNICAL SKILLS

Languages: Java, C and C++, Python, JavaScript and TypeScript, SQL, HTML and CSS

Operating Systems: Linux and Unix, Embedded Linux, Windows

CS Foundations: Object-Oriented Programming (OOP), Data Structures & Algorithms, Complexity Analysis, System Design, REST APIs, Authentication with OAuth2 and JWT, Agile and Scrum

Embedded & Firmware: Embedded C, ARM Cortex-M and Cortex-M7, STM32, PIC microcontrollers, SafeRTOS, Qt and QML, Yocto, SPI, I2C, UART, RS-422, GPIO, PWM, watchdog timers, motor control, state machines

Databases: MySQL, MongoDB, Redis

Frameworks & Tools: Spring Boot, Node.js, Express, Next.js, React, Docker, Kubernetes, Jenkins, GitLab CI and CD, AWS, GCP, Kafka, gRPC, Kong Gateway, Prometheus, Git

Scripting & Automation: Python, Shell scripting, CI and CD pipelines, automated testing